



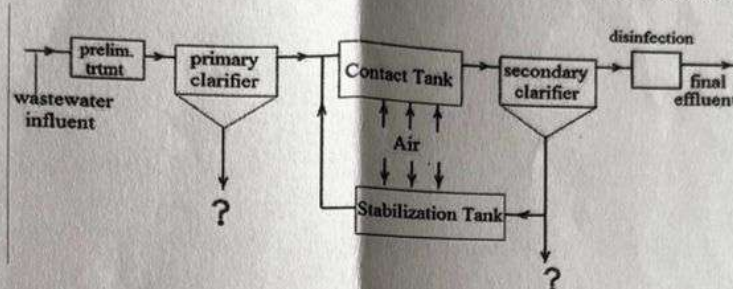
Department of Biosciences and Bioengineering
BT 416
Bioenvironmental Engineering

Date: 08/05/2026

End Semester Examination
Answer ALL questions

Total Marks: 50

1. In the following figure, sedimentation basins perform the two-fold function of producing both a clarified water product and a concentrated slurry (sludge) product. Therefore, sedimentation vessels can be called both as a clarifier and thickener

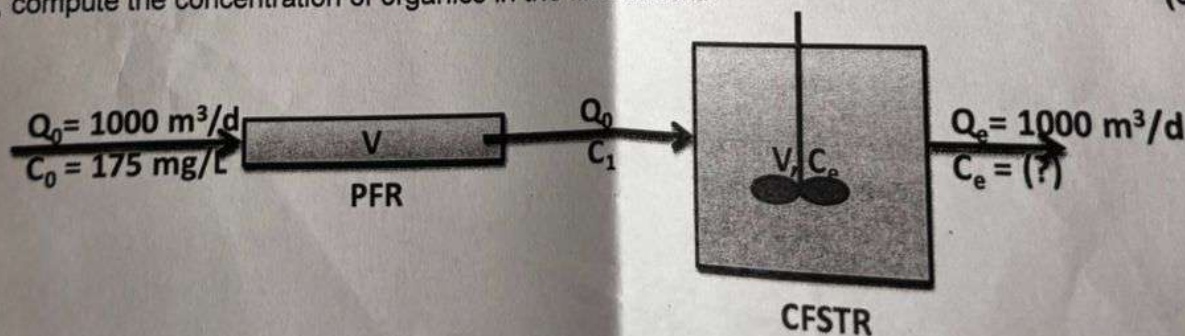


Flow sheet showing a typical wastewater treatment plant

Answer the following questions:

(5 marks)

- What is the purpose of the primary clarifier in this flow sheet?
 - Though the primary clarifier performs well, what use is the secondary clarifier shown in the figure?
 - What does '?' in the flow sheet indicate? Mention their sources.
 - The settling characteristic of sludge is often measured in terms of sludge volume index (SVI), which is defined as the ratio of settled volume of sludge (ml/L) to the concentration of suspended solids (mg/L). SVI of a one-litre sample containing 3000 mg/L of suspended solids is tested using a laboratory-scale sedimentation setup. If the volume occupied by the settled solids in the sample is 300 ml in 30 minutes, determine the value of SVI in ml/g.
2. How does anaerobic oxidation of organics differ from aerobic oxidation? What are the main advantages and disadvantages of anaerobic treatment over aerobic treatment? Mention the various classes of microorganisms involved in anaerobic treatment, and describe the reactions they carry out. (5 marks)
3. With the help of schematics, describe salient features along with advantages and disadvantages of the following wastewater treatment systems: (3 × 5 = 15 marks)
(a) Facultative pond (b) Rotating biological contactor reactor and (c) Upflow anaerobic sludge blanket reactor.
4. A wastewater treatment system, as shown in the following figure, comprising of a PFR followed by a CFSTR in series is to be designed for treating wastewater with a flow rate of $1000 \text{ m}^3/\text{d}$ and containing 175 mg/L organics. If the volume of each reactor is 2500 m^3 and the reaction follows first order kinetics with a k value of 0.8 d^{-1} , compute the concentration of organics in the final effluent. (5 marks)



5. An activated sludge process is to be designed to treat 10 million litres per day (MLD) of domestic wastewater, with an MCRT of 10 days and an HRT of 6 hours. Assuming BOD_5 at 20 °C as 175 mg/L in influent to the aeration tank, sludge wasting flow equal to 70 m³/d and the returned sludge concentration equal to 8000 mg/L, determine the concentration of MLVSS (in mg/L) to be maintained in the aeration tank to achieve an effluent BOD_5 of 30 mg/L.

Also, determine the recirculation ratio at which the plant should be operated.

Given: $k_d = 0.06 \text{ d}^{-1}$, $Y = 0.6$ and $X_e = 0$.

(5 marks)

6. A low-rate trickling filter has a diameter of 40 m and a depth of 1.5 m. Using the Eckenfelder equation, determine treatment efficiency (in %) of the trickling filter for a wastewater flow of 2000 m³/d and 200 mg/L BOD_5 . Given the values of removal rate constant (k) = 1.90 d⁻¹ and filter media constant (n) = 0.67. (5 marks)

7. A meat processing plant uses an anaerobic contact process, maintained at 35 °C temperature, to treat the raw wastewater with a flow rate of 200 m³/d and an initial organic concentration of 3000 mg/L. The effluent from the anaerobic treatment process contains 461 mg/L of organics, and the observed biomass yield (Y_{obs}) value in the process is determined to be 0.027. Estimate the biomass produced in kg/d, nutrient (N and P) requirement in kg/d and methane production rate in m³/d. (5 marks)

8. A packed bed reactor containing immobilized biomass in the form of biofilm is used to treat wastewater containing organics of initial concentration (S_0) = 2 kg/m³ in an upflow mode at the flow rate, $F = 1 \text{ m}^3/\text{hr}$. The substrate conversion rate (r_s) is given by the expression:

$$r_s = \frac{kXS}{K_s + S}$$

where k , the maximum specific rate of conversion, is 0.5 h⁻¹, X , the biomass concentration, is 10 kg/m³, half-saturation constant, K_s , is 0.2 kg/m³ and S is the concentration of organics present in the wastewater.

If the reactor has a specific surface area of 100 m²/m³ and a cross-sectional area of 4 m², calculate the required height (in m) of the column to achieve an effluent organic concentration (S) = 0.03 kg/m³. Given the values of biofilm thickness = 0.02 cm and the value of the effectiveness factor (η) = 0.8 in the immobilized bioreactor system. (5 marks)